

5(a)(i)	<i>three from:</i> 1 papain higher optimum temperature than ficin / ora or temperature is higher for papain for percentage of maximum activity of 100 A 100% activity / maximum activity / peak of activity or (100% of maximum activity) papain 60 °C v ficin 45 °C ; 2 papain (overall) less steep increase / AW, (to peak) / ora ; 3 papain steeper decline / AW (after peak) / ora ; <i>for mp 2 and 3 allow ecf for a ref. to time, e.g. rapidly / slowly</i> 4 papain lower activity at all temperatures between 20 °C and, 54 °C / 55 °C ; 5 papain higher activity at all temperatures above, 54 °C / 55 °C ; 6 ref. to ficin more active over a greater range of temperatures ; A ref. to activity at temperatures around 100% maximum 7 comparative, extracted / manipulated, data to support mp, 2–6 ;	3
5(a)(ii)	ficin, as higher activity at 37 °C / body temperature ;	1
5(b)	<i>one from:</i> (may be able to) obtain more product (per unit time) ; A can use in a continuous system (may be) more, thermostable / tolerant to pH changes / AW ; A idea that enzyme less likely to denature product, not / less, contaminated with enzyme ; A less downstream processing longer shelf-life / AW ; enzyme, easier to recover / can be reused ; AVP ; e.g. ficin may be, costly / difficult, to extract	1
5(c)(i)	extracellular (enzyme) ;	1

Question	Answer	Marks
5(c)(ii)	<i>any one valid example:</i> e.g. collagen / elastin ;	1

Question	Answer	Marks
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